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## Electrical connector having four terminal modules and a pair of metallic side plates fixing the four terminal modules


### Abstract

An electrical connector including: an insulative housing; four terminal modules received in the insulative housing and each having a row of terminals integrally formed with plural plastic fixing members, each terminal defining a contact portion, a tail portion opposite to the contact portion, and a holding portion therebetween; and a pair of metallic side plates fixing the four terminal modules together, each metallic side plates having plural holes, each plastic fixing member including a pair of positioning protrusions at two ends thereof to be received within corresponding holes, wherein the positioning protrusions respectively extend through the corresponding holes and are heat riveted to the metallic side plates.



**Inventors:** **Fu; Dai-Pei (Kunshan, CN), Hsu; Chun-Hsiung (New Taipei, TW)**

**Applicant:** **FOXCONN (KUNSHAN) COMPUTER CONNECTOR CO., LTD.** (Kunshan, CN); **FOXCONN INTERCONNECT TECHNOLOGY LIMITED** (Grand Cayman, KY)



**Family ID:** **1000008777997**

**Assignee:** **FOXCONN (KUNSHAN) COMPUTER CONNECTOR CO., LTD. (Kunshan, CN); FOXCONN INTERCONNECT TECHNOLOGY LIMITED (Grand Cayman, KY)**

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*Primary Examiner:* Dinh; Phuong K

## Background/Summary

BACKGROUND OF THE INVENTION
Field of the Invention
(1) The present invention relates generally to an electrical connector, and particularly to an electrical connector including four terminal modules secured by a pair of metal side plates.
Description of Related Arts
(2) US Patent Application Publication No. 20210328384 discloses an electrical connector comprising a four terminal modules and a pair of metallic side plate for fixes the four terminal modules, each metallic side plates forms a plurality of apertures therein, and each terminal module includes positioning protrusions to be received within the corresponding apertures. However, the positioning protrusions and the aperture create an interference fit/compression fit force, and it requires high precision to match, and more complicated to assemble and the fix is not stable.
(3) An improved electrical connector is desired.
SUMMARY OF THE INVENTION
(4) The objective of the present invention is to provide an electrical connector including four terminal modules that is easy to assemble.
(5) To achieve the above object, an electrical connector comprises: an insulative housing; a first terminal module and a second terminal module arranged vertically in the insulative housing, and a third terminal module and a fourth terminal module arranged vertically in the insulative housing, the first terminal module including a first row of terminals integrally formed with a plurality of plastic fixing members extending in a transverse direction perpendicular to the vertical direction, the second terminal module including a second row of terminals integrally formed with a plurality of transversely extending plastic fixing members via insert-molding, the third terminal module including a third row of terminals integrally formed with a plurality of transversely extending plastic fixing members via insert-molding, the fourth terminal module including a fourth row of terminals integrally formed with a plurality of transversely extending plastic fixing members via insert-molding, each terminal defining a contact portion matched with a mating connector, a tail portion opposite to the contact portion, and a holding portion therebetween; and a pair of metallic side plates fixing the four terminal modules together, each metallic side plates having a plurality of holes, each plastic fixing member including a pair of positioning protrusions at two ends thereof in the transverse direction to be received within corresponding holes, respectively, wherein the positioning protrusions respectively extend through the corresponding holes and are heat riveted to the metallic side plates.
(6) Compared to prior art, in the electrical connector of the present invention, the positioning protrusions of each terminal module are respectively passed through the corresponding holes and secured to the metallic side plate by heat riveting. Therefore, the assembly is simple and stable and the requirements for matching accuracy are lower.

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## Description

BRIEF DESCRIPTION OF THE DRAWING
(1) FIG. **1** is a perspective view of an electrical connector according to the first invention;
(2) FIG. **2** is another perspective view of the electrical connector of FIG. **1**;
(3) FIG. **3** is an exploded perspective view of the electrical connector of FIG. **1**;
(4) FIG. **4** is another exploded perspective view of the electrical connector of FIG. **3**;
(5) FIG. **5** is a further exploded perspective view of the electrical connector of FIG. **3**;
(6) FIG. **6** is another exploded perspective view of the electrical connector of FIG. **5**;
(7) FIG. **7** is an exploded perspective view of the electrical connector of FIG. **6** without showing the insulative housing;
(8) FIG. **8** is a side view of the electrical connector of FIG. **7** without showing the insulative

housing;
(9) FIG. **9** is a side view of the terminal module of the electrical connector of FIG. **7**;
(10) FIG. **10** is perspective view of the terminal module of the electrical connector of FIG. **7**;
(11) FIG. **11** is another perspective view of the terminal module of the electrical connector of FIG. **10**;
(12) FIG. **12** is a exploded perspective view of the terminal module of FIG. **10**;
(13) FIG. **13** is another perspective view of the terminal module of FIG. **12**; and
(14) FIG. **14** is a cross-sectional view of the electrical connector of FIG. **1**.
DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENT
(15) Referring to FIGS. **1-14**, an electrical connector **100** in accordance with the present invention is shown. The electrical connector **100** can be inserted and matched with a mating connector (not shown) and includes an insulative housing **10** cooperating with an insulative upper cover **30** to commonly receive a terminal module **20** therein.
(16) The terminal module **20** includes a first terminal module **210**, a second terminal module **220**, a third terminal module **230** and a fourth terminal module **240**. The first terminal module **210** and the third terminal module **230** are stacked with each other in the vertical direction. The second terminal module **220** and the fourth terminal module **240** are stacked with each other in the vertical direction.
(17) The first terminal module **210** includes a first row of terminals **212** arranged along the transverse direction integrally formed with a plurality of transversely extending plastic fixing members **215** via insert-molding. Each plastic fixing member **215** includes an insulative primary part **214** and a conductive secondary part **216**. Similarly, the second terminal module **220** includes a second row of terminals **222** integrally formed with a plurality of transversely extending plastic fixing members **225** via insert-molding. Each plastic fixing member **225** includes an insulative primary part **224** and a conductive secondary part **226**. The third terminal module **230** includes a third row of terminals **232** integrally formed with a plurality of transversely extending plastic fixing members **235** via insert-molding. Each plastic fixing member **235** includes an insulative primary part **234** and a conductive secondary part **236**. The fourth terminal module **240** includes a fourth row of terminals **242** integrally formed with a plurality of transversely extending plastic fixing members **245** via insert-molding. Each of said plastic members **245** includes an insulative primary part **244** and a conductive secondary part **246**. Each terminal includes a contacting section **271** matched with the mating connector, a tail portion **272** opposite to the contact portion, and a holding portion **273** therebetween. Each row of terminals includes a plurality of differential-pair signal terminals and a plurality of ground terminals alternately arranged with each other in the transverse direction. The ground terminals are further joined together by the corresponding conductive secondary parts. The width of the signal terminal is smaller than the width of the ground terminal.
(18) Each of the terminals includes a covered portion **201** arranged in the corresponding plastic fixing member and a free portion **202** exposed to air, the length of the free portion **202** is longer than that of the covered portion **201**. The plastic fixing member **215** has an opening **217** that exposes the corresponding signal terminal to air. Specifically, take the first terminal module **210** as an example, the insulative primary part **214** has an opening **217** that exposes the corresponding signal terminal to air, and the conductive secondary part **216** has a notch **218** aligned with the opening **217**. The first row of terminals **212** and the second row of terminals **222** form a first mating port **1**, and the third row of terminals **232** and the fourth row of terminals **242** form a second mating port **2**. The first mating port **1** and the second mating port **2** are arranged along a front-to-back direction. Because the ground terminal is wider than the signal terminal, the plastic fixing member near the contact portion **271** on the signal terminal extends forward beyond the portion on the ground terminal to increase the contact normal force of the signal terminal. The length of the free portion **202** of at least one of the first row of terminals **212**, the second row of

terminals **222**, the third row of terminals **232**, and the fourth row of terminals **242** is at least three times the length of the covered portion **201**. Specifically, the length of the free portion **202** of each terminal of the first row is of terminals **212** is at least twice the length of the covered portion **201**. The length of the free portion **202** of each terminal of the second row of terminals **222** is at least 1.5 times the length of the covered portion **201**. The length of the free portion **202** of each terminal of the third row of terminals **232** is at least three times the length of the covered portion **201**. The length of the free portion **202** of each terminal of the fourth row of terminals **242** is at least twice the length of the covered portion **201**.

(19) A pair of metallic side plates **40** are located by two opposite sides of the terminal module **20**, respectively. The metallic side plate **40** fixes the first terminal module **210**, the second terminal module **220**, the third terminal module **230** and the fourth terminal module **240** together. Each metallic side plates **40** forms a plurality of holes **402** therein. Correspondingly, each plastic fixing member includes a pair of positioning protrusions **204** at two opposite ends to be received within the corresponding holes **402**, respectively, so as to secure the whole terminal module **20** together. The positioning protrusions **204** secured to the metallic side plate **40** by heat riveting.

(20) Each metallic side plate **40** forms a notch **408** is aligned with the contact portion **271** of the terminals of the third row of terminals **232** the fourth row of terminals **242** in the transverse direction. The plastic fixing member **215** of the first terminal module **210** forms a plurality of grooves **2151** for respectively receive the contact portions **271** of the corresponding terminals of the third row of terminals **232**. The plastic fixing member **225** of the second terminal module **220** forms a plurality of grooves **2251** to respectively receive the contact portions **271** of the corresponding terminals of the fourth row of terminals **242**.

(21) A front region of the insulative hosing **10** forms a mating slot **110**, and upper and lower rows of passageways **112** located two sides of the mating slot **110** in the vertical direction wherein the upper row of passageways **112** respectively receive the contact portion **271** of the first row of terminals **212**, and the lower passageways **112** respectively receive the contact portion **271** of the second row of terminals **222**. The insulative housing **10** includes an upper wall **11**, a lower wall **12** and a pair of side walls **13**. The upper wall **11** includes a flat horizontal portion **101** at the front and a raised portion **103** at the rear of the horizontal portion **101**. An elongated opening **105** is formed between the horizontal portion **101** and the raised portion **103**. A rear region of the insulative housing **10** forms a pair of upper channels **113** and a pair of lower channels **114** in two opposite side walls **13**. Correspondingly, the positioning protrusions **204** are arranged in upper and lower rows to be respectively received within the upper channels **113** and lower channels **114**. Therefore, the assembled terminal module **20** can be inserted into the insulative housing **10** from the rear side forwardly along the front-to-back direction. A pair of stopping blocks **109** is formed on the inner faces of the side walls **13** to be received within the corresponding notches **408** of the corresponding metallic side plate **40**.

(22) The insulative upper cover **30** includes a slot **352** extending in the transverse direction to receive one plastic fixing member **215** therein, and a pair of guide ribs **354** received within the upper channel **113**, and a protrusion **358** received within the locking space **119** in the upper channel **113**. Therefore, the insulative upper cover **30** is combined with the terminal module **20** as a whole, forwards along the front-to-back direction, and are inserted into the insulative housing **10** from the rear side of the insulative housing **10**.

(23) The electrical connector **100** of the present invention exposes the signal terminals to air as much as possible to adjust the characteristic impedance of the signal terminals, thereby improving the signal transmission rate of the connector. Moreover, the electrical connector **100** of the present invention fixes the terminal module **20** as a whole through the side plates **40** on the two sides, so that the structure of the electrical connector **100** is more compact, and the assembly is simple and convenient. The electrical connector **100** of the present invention conforms to the specification of QSFP-DD, which defines 8 transmission channels and 8 reception channels, and the signal

transmission rate of each channel can reach 100 Gbps. Of course, the present invention can also be applied to high-speed electrical connectors such as SFP-DD, SFP, OSFP, etc.

(24) The above-mentioned embodiments are only preferred embodiments of the present invention, and should not limit the scope of the present invention, any simple equivalent changes and modifications made according to the claims of the present invention and the contents of the description should still belong to the present invention.

## Claims

1. An electrical connector comprising: an insulative housing; a first terminal module and a second terminal module arranged vertically in the insulative housing, and a third terminal module and a fourth terminal module arranged vertically in the insulative housing, the first terminal module including a first row of terminals integrally formed with a plurality of plastic fixing members extending in a transverse direction perpendicular to a vertical direction, the second terminal module including a second row of terminals integrally formed with a plurality of transversely extending plastic fixing members via insert-molding, the third terminal module including a third row of terminals integrally formed with a plurality of transversely extending plastic fixing members via insert-molding, the fourth terminal module including a fourth row of terminals integrally formed with a plurality of transversely extending plastic fixing members via insert-molding, each terminal defining a contact portion matched with a mating connector, a tail portion opposite to the contact portion, and a holding portion therebetween; and a pair of metallic side plates fixing the four terminal modules together, each metallic side plates having a plurality of holes, each plastic fixing member including a pair of positioning protrusions at two ends thereof in the transverse direction to be received within corresponding holes, respectively; wherein the positioning protrusions respectively extend through the corresponding holes and are heat riveted to the metallic side plates.
2. The electrical connector as claimed in claim 1, wherein each of the four rows of terminals includes a covered portion held in corresponding plastic fixing members and a free portion exposed to air, a length of the free portion of at least one terminal of the first row of terminals, the second row of terminals, the third row of terminals, and the fourth row of terminals is at least three times a length of an associated covered portion.
3. The electrical connector as claimed in claim 2, wherein the first row of terminals and the second row of terminals form a first mating port, and the third row of terminals and the fourth row of terminals form a second mating port, and the first mating port and the second mating port are arranged along a front-to-back direction.
4. The electrical connector as claimed in claim 2, wherein the length of the free portion of each terminal of the third row of terminals is at least three times the length of the covered portion.
5. The electrical connector as claimed in claim 2, wherein the length of the free portion of each terminal of the first row of terminals is at least twice the length of the covered portion, the length of the free portion of each terminal of the second row of terminals is at least twice the length of the covered portion, and the length of the free portion of each terminal of the fourth row of terminals is at least twice the length of the covered portion.
6. The electrical connector as claimed in claim 2, wherein each row of terminals includes plural pairs of signal terminals capable of transmitting differential signals and plural ground terminals alternately arranged with the plural pairs of signal terminals, each of the ground terminals is wider than each of signal terminals of the plural pair of signal terminals, a portion of the plastic fixing member near the contact portion on the signal terminal extends forward beyond an another portion of the plastic fixing member near the contact portion on the ground terminal to increase the contact normal force of the signal terminal.
7. The electrical connector as claimed in claim 6, wherein each plastic fixing member includes an insulative primary part and a conductive secondary part, the conductive secondary part

mechanically and electrically connects corresponding ground terminals together.
8. The electrical connector as claimed in claim 7, wherein the plastic fixing member has an opening that exposes a corresponding signal terminal to air.
9. The electrical connector as claimed in claim 8, wherein the insulative primary part has an opening that exposes the corresponding signal terminal to air, and the conductive secondary part has a notch aligned with the opening.
10. The electrical connector as claimed in claim 1, wherein a rear region of the insulative housing forms a pair of upper channels and a pair of lower channels in two opposite side walls thereof, and the positioning protrusions are arranged in upper and lower rows to be respectively received within the upper channels and lower channels.
11. The electrical connector as claimed in claim 1, further comprising an insulative upper cover including a slot extending in the transverse direction to receive one plastic fixing member therein, and wherein the insulative upper cover and the terminal modules are inserted into the insulative housing from the rear side of the insulative housing.
12. The electrical connector as claimed in claim 1, wherein the insulative housing includes an upper wall, a lower wall, and a pair of side walls, the upper wall includes a flat horizontal portion at the front and a raised portion at the rear of the flat horizontal portion, and an elongated opening is formed between the horizontal portion and the raised portion.